# Requests Responses and JSON

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

## One connected JSON and interface record.

Translate values, classify failures, guard structure and state what parsing cannot prove.

**Keep one habit:** Preserve exact fields, types, ranges, output behavior and factual-evidence boundaries.

MAIN LESSON · SECTION 1

## Recall: reconnect earlier ideas

> Retrieve before opening notes.

### Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Tools and APIs. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from User Need and AI Contribution. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Verify Numbers and Dates. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

**My remembered idea, example and correction**

**My remembered idea, example and correction**

▼ **Earlier wording, alternatives and picture transcripts**

**Tools and APIs**

**User Need and AI Contribution**

**Verify Numbers and Dates**

## W1 · Translate representations

> Keep JSON and Python spellings separate.

### Your evidence record

- Activity contract: exactly name nonempty text and minutes exact integer 0..60; reject bool, extra/duplicate fields, malformed JSON and non-object top levels. Rejection output accepted=false,minutes=null; accepted minutes 0 is legitimate zero. JSON null→Python None; Python True→JSON true. Separate lunch contract uses class_code/portions 0..30. Parsing, schema validity and observation truth are separate.

- What Python value does **JSON** null become? What **JSON** value represents Python True?

> **My reasoning and evidence**

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MAIN LESSON · SECTION 13

## W2 · Separate syntax from type errors

> State all missing fields as well as the intended contrast.

## Your evidence record

- Activity contract: exactly name nonempty text and minutes exact integer 0..60; reject bool, extra/duplicate fields, malformed JSON and non-object top levels. Rejection output accepted=false,minutes=null; accepted minutes 0 is legitimate zero. JSON null→Python None; Python True→JSON true. Separate lunch contract uses class_code/portions 0..30. Parsing, schema validity and observation truth are separate.

- Classify {"minutes": "5"} and {minutes: 5} as valid **JSON** text or malformed **JSON**.

> **My reasoning and evidence**

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MAIN LESSON · SECTION 14

# W3 · Guard the top-level value

> Check dictionary structure before field access.

## Your evidence record

- Activity contract: exactly name nonempty text and minutes exact integer 0..60; reject bool, extra/duplicate fields, malformed JSON and non-object top levels. Rejection output accepted=false,minutes=null; accepted minutes 0 is legitimate zero. JSON null→Python None; Python True→JSON true. Separate lunch contract uses class_code/portions 0..30. Parsing, schema validity and observation truth are separate.

- A parser accepts a **JSON** list then crashes on .get. What check belongs before field access?

**My reasoning and evidence**

MAIN LESSON · SECTION 16

## W5 · Limit what parsing proves

> Do not infer source truth from a successful parse.

### Your evidence record

- Activity contract: exactly name nonempty text and minutes exact integer 0..60; reject bool, extra/duplicate fields, malformed JSON and non-object top levels. Rejection output accepted=false,minutes=null; accepted minutes 0 is legitimate zero. JSON null→Python None; Python True→JSON true. Separate lunch contract uses class_code/portions 0..30. Parsing, schema validity and observation truth are separate.

- A **JSON** record parses successfully. Does that prove its **observation** is correct?

**My reasoning and evidence**

MAIN LESSON · SECTION 4

# W4 · Specify the new label/count interface independently

> Keep one first attempt before hints or feedback.

## Read and explain independently

Specify your choices explicitly for missing or extra fields and the output on rejection. The task supplies field meanings and bounds, but no real box inventory. Treat acceptance as a contract decision rather than evidence that a box was physically counted.

Define a **JSON contract** for {"label": "Box", "count": 4}: nonempty text label, integer count 0 through 10. Give a valid case, a wrong-type case and a missing-field case with expected acceptance.

> **My independent explanation, trace and limits**

## Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

**Independent case: Specify independently**

## Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

### Your evidence record

Use a stable **request/response contract** between **interface**, retriever, model, and verifier.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

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MAIN LESSON · SECTION 17

## Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

### Your evidence record

| Evidence | My record |
|---|---|
| Worked alone / hint / model | |
| First step I repaired and why | |
| New case checked later; date and result | |
| What I can now explain without notes | |

| Evidence | My record |
| --- | --- |
| One remaining question | |

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

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MAIN LESSON · SECTION 10

## Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

### Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

# Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

## Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

### Use this when helpful

Use W2: Separate failures for **JSON** and **interface contract**s. Record the answer once; the lesson and workbook refer to the same attempt.

| Given inputs | Procedure I chose | Middle steps | Result and check |
|---|---|---|---|
| | | | |

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

## Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

### Use this when helpful

Use the worked case for **JSON** and **interface contract**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

## Reference · Original learning target

> Use this target to inspect an explanation.

### Use this when helpful

**Request** id=7 returns **status**=200 and {"id":8,"pages":90}. Explain why a successful **status** is insufficient and specify the handling.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

## Extra practice · A complete case

> Try the question before checking the worked explanation.

### Use this when helpful

**Request** is for B7; **response** is {"id":"B7","pages":120}. Another **response** has malformed **JSON**.

What checks precede use?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

## Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

### Use this when helpful

An API returns {"ok":false,"error":"not_found"}. Your app tries to display **response**.title. What should it do instead?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

**Week 22 Test and reflect**

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

## Extra practice · Design a revealing test

> Predict before tracing or testing.

### Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe **response** before testing.

| Test record | My evidence |
| --- | --- |
| Changed input or condition | |
| Expected answer or fallback | |
| Observed result | |
| Why the result occurred | |

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

# Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

## Use this when helpful

**Request** id=7 returns **status**=200 and {"id":8,"pages":90}. Explain why a successful **status** is insufficient and specify the handling.

**My explanation after practice**

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

# Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

## Use this when helpful

Define a **JSON contract** for {"label": "Box", "count": 4}: nonempty text label, integer count 0 through 10. Give a valid case, a wrong-type case and a missing-field case with expected acceptance.

> **My reasoning and evidence**

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

# Extra practice · Use the worked picture

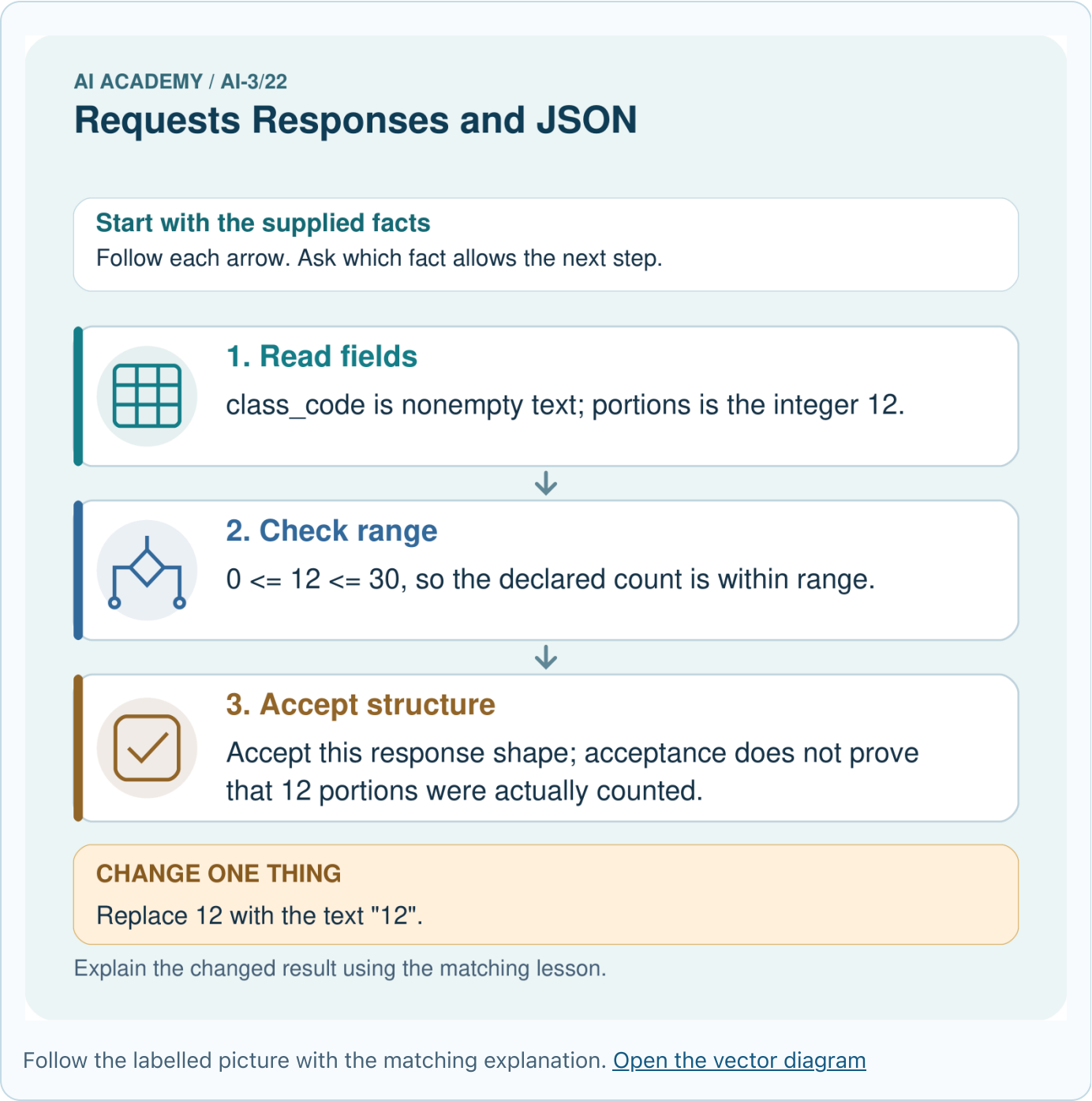
> Keep the explanation and the support you used.

## Use this when helpful

A lunch-count **response** must satisfy a declared **JSON contract**.

The toy **response** has exactly class_code text and portions integer 0..30. Example: {"class_code":"R8","portions": 12}.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.




Follow the labelled picture with the matching explanation. Open the vector diagram

Would converting the text silently prove the source count is correct? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

**My picture-based explanation and changed-case prediction**

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

# Reference · Project portfolio record

> Extend your main project response when helpful.

## Use this when helpful

Project: Verified Study Buddy

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

| Evidence to keep | My record |
| --- | --- |
| This week's input and deciding step | |
| Prediction and actual result (or labelled paper trace) | |
| One change since the previous checkpoint and why | |
| One error or limit and the next test | |

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

# Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

## First supported question

- A lunch-count **response** must satisfy a declared **JSON contract**.
- Which labelled step matches this detail? Accept this **response** shape; acceptance does not prove that 12 portions were actually counted.
- 1. Accept structure

- 2. Read fields
- 3. Check range
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.
- After your attempt, compare with the supported explanation: Accept structure Accept this **response** shape; acceptance does not prove that 12 portions were actually counted. Read the picture in order.

  Accept this **response** shape; acceptance does not prove that 12 portions were actually counted. The arrows show which recorded **input** or intermediate result each decision uses.

  Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

### Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> | |
> |---|
> | |
> | |
> | |

▼ **Check the first explanation**

### Practice 2: explain the deciding evidence

## Changed-case question

- Changed-case practice: Replace 12 with the text "12". Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.

- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. Reject the **response** under this strict **contract**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.
- After your attempt, compare with the supported explanation: Reject the **response** under this strict **contract**. Digits inside quotes represent text, whereas portions requires an integer.

> **My selected conclusion, deciding fact and explanation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Check the changed case**

**Repair a missing building block**

## Repair and save your work

Review Tools and APIs (ai-3/21). Goal: Explain the **mechanism** and terminology of tools and api **contract**s. Short practice: Does a valid C7 ID permit the extra lending action?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review User Need and AI Contribution (ai-3/19). Goal: Explain the **mechanism** and terminology of capstone requirements and corpus. Short practice: What non-AI baseline would reveal whether this helper adds value?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Verify Numbers and Dates (ai-3/16). Goal: Verify a numeric claim using its source, denominator and units. Short practice: Which part can be corrected immediately from existing **evidence**, and which needs follow-up?. Return to this week and explain what you

changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

**Predict, run and debug**

**Keep your completed work**

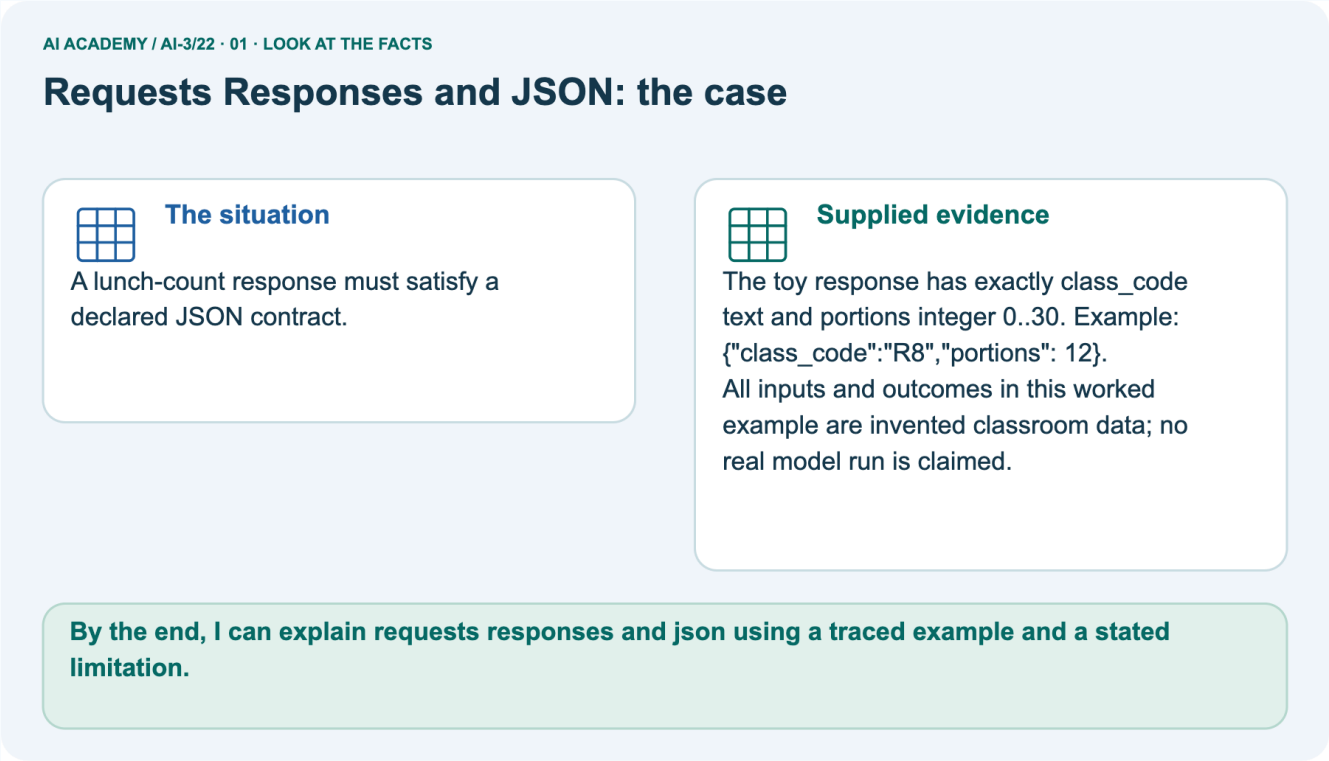
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

## Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

## 01. Requests Responses and JSON: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

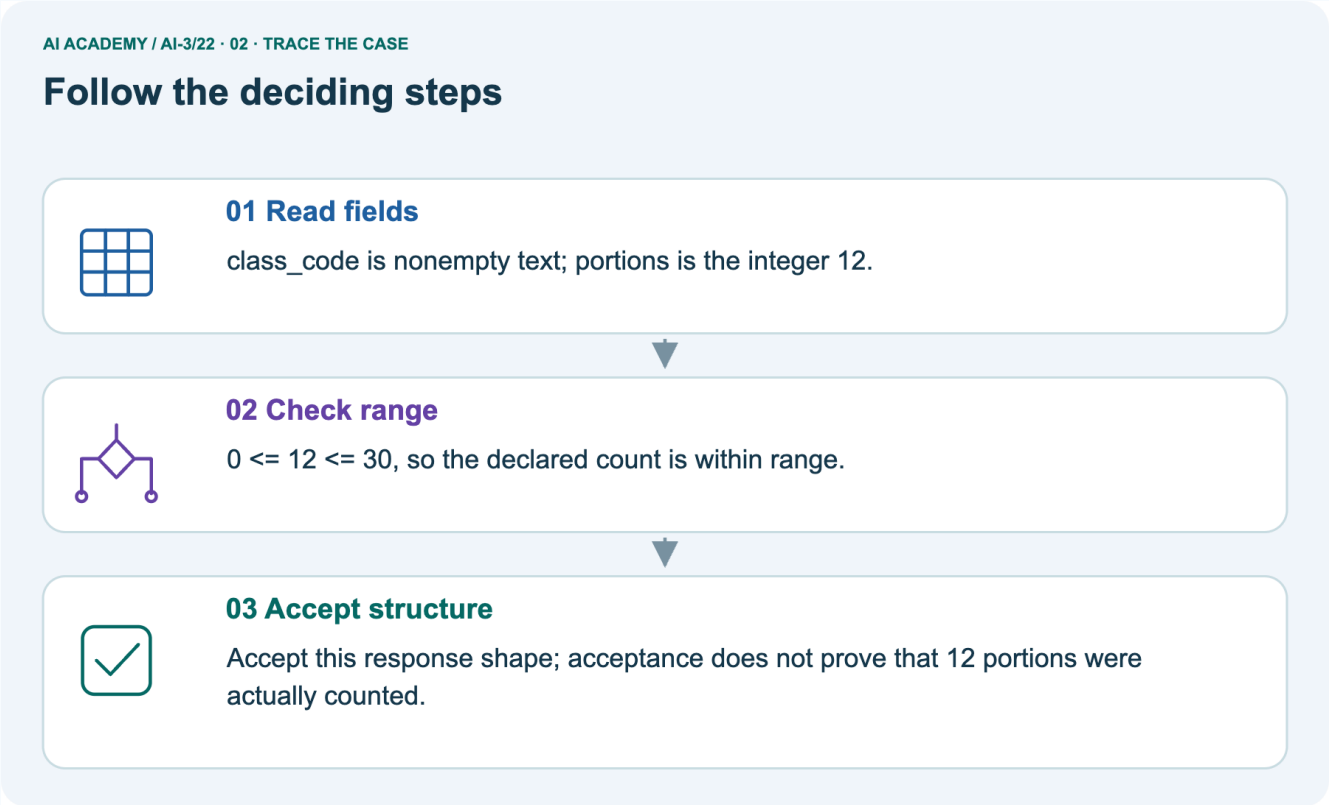
A lunch-count **response** must satisfy a declared **JSON contract**. Supplied facts: The toy **response** has exactly class_code text and portions integer 0..30. Example: {"class_code":"R8","portions": 12}.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

### My picture explanation, deciding evidence and one limit

## 02. Follow the deciding steps




Follow the labelled picture with the matching explanation. Open the vector diagram

Read fields: class_code is nonempty text; portions is the integer 12. → Check range: 0 <= 12 <= 30, so the declared count is within range. → Accept structure: Accept this **response** shape; acceptance does not prove that 12 portions were actually counted.

Point to each numbered stage and explain which recorded **input** it uses.

### My picture explanation, deciding evidence and one limit

## 03. Follow the evidence

AI ACADEMY / AI-3/22 · 03 · INSPECT THE LESSON MODEL

### Follow the evidence

| Step or case | Evidence | Meaning |
| --- | --- | --- |
| Input text | JSON object | Parse with json.loads |
| Python value | Dictionary fields | Check schema |
| Output text | accepted and minutes | Serialize with json.dumps |

Read each row from left to right and connect the evidence to the stated rule.

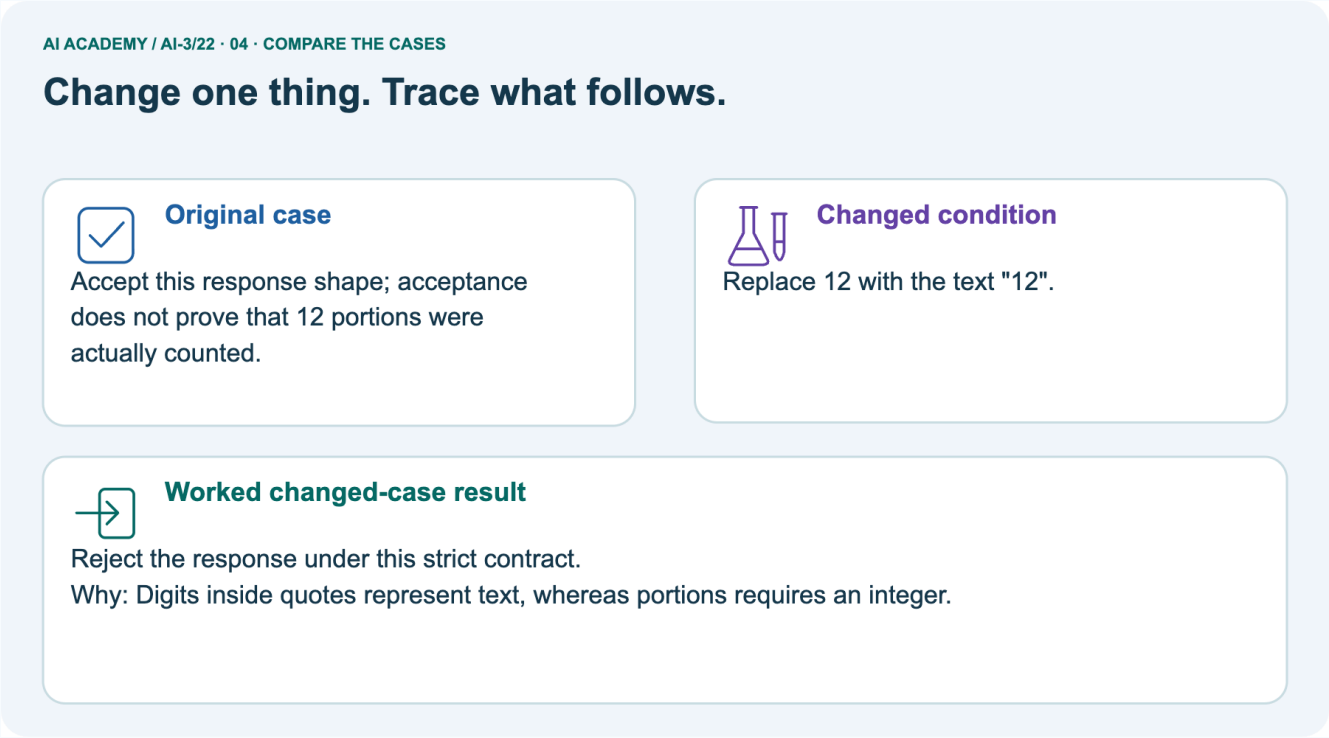
Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. **Input** text | **JSON object** | Parse with **json**.loads. Python value | Dictionary fields | Check schema. **Output** text | accepted and minutes | Serialize with **json**.dumps

Read the labels and values. Explain the relationship the model shows.

### My picture explanation, deciding evidence and one limit

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

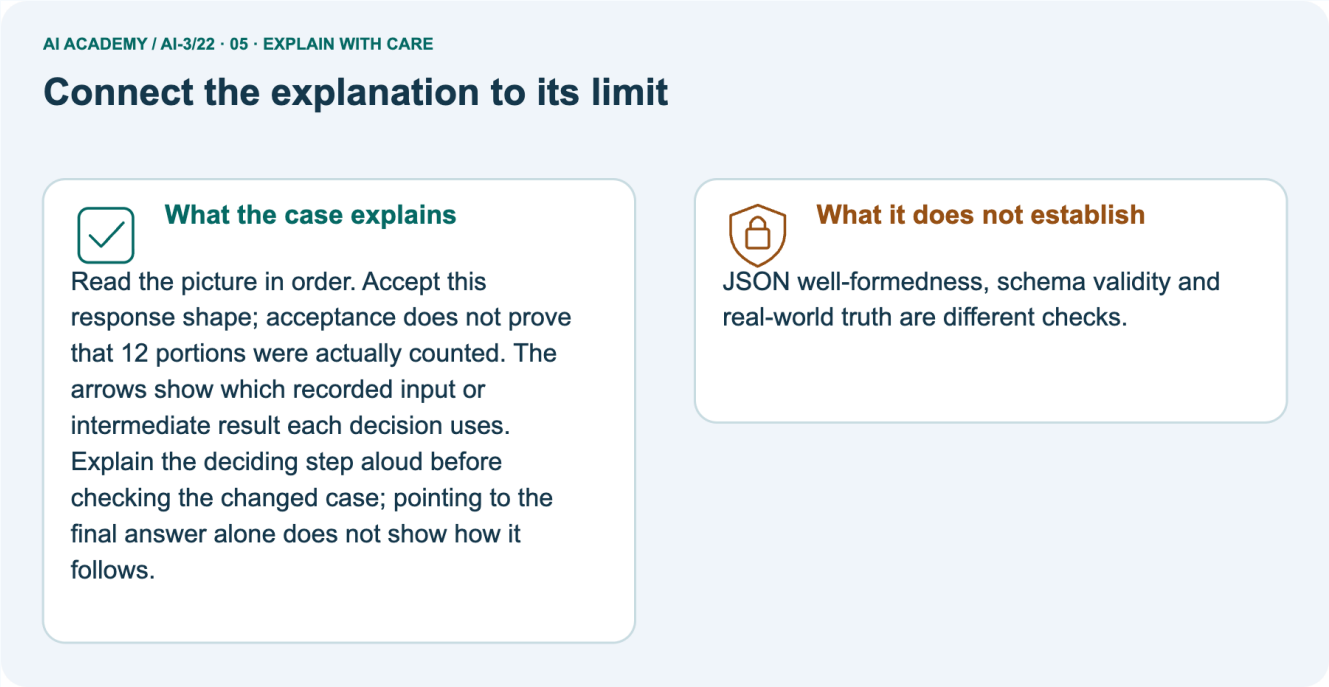
Original case: Accept this **response** shape; acceptance does not prove that 12 portions were actually counted. Changed condition: Replace 12 with the text "12". Expected result: Reject the **response** under this strict **contract**.

Reason: Digits inside quotes represent text, whereas portions requires an integer.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

### My picture explanation, deciding evidence and one limit

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

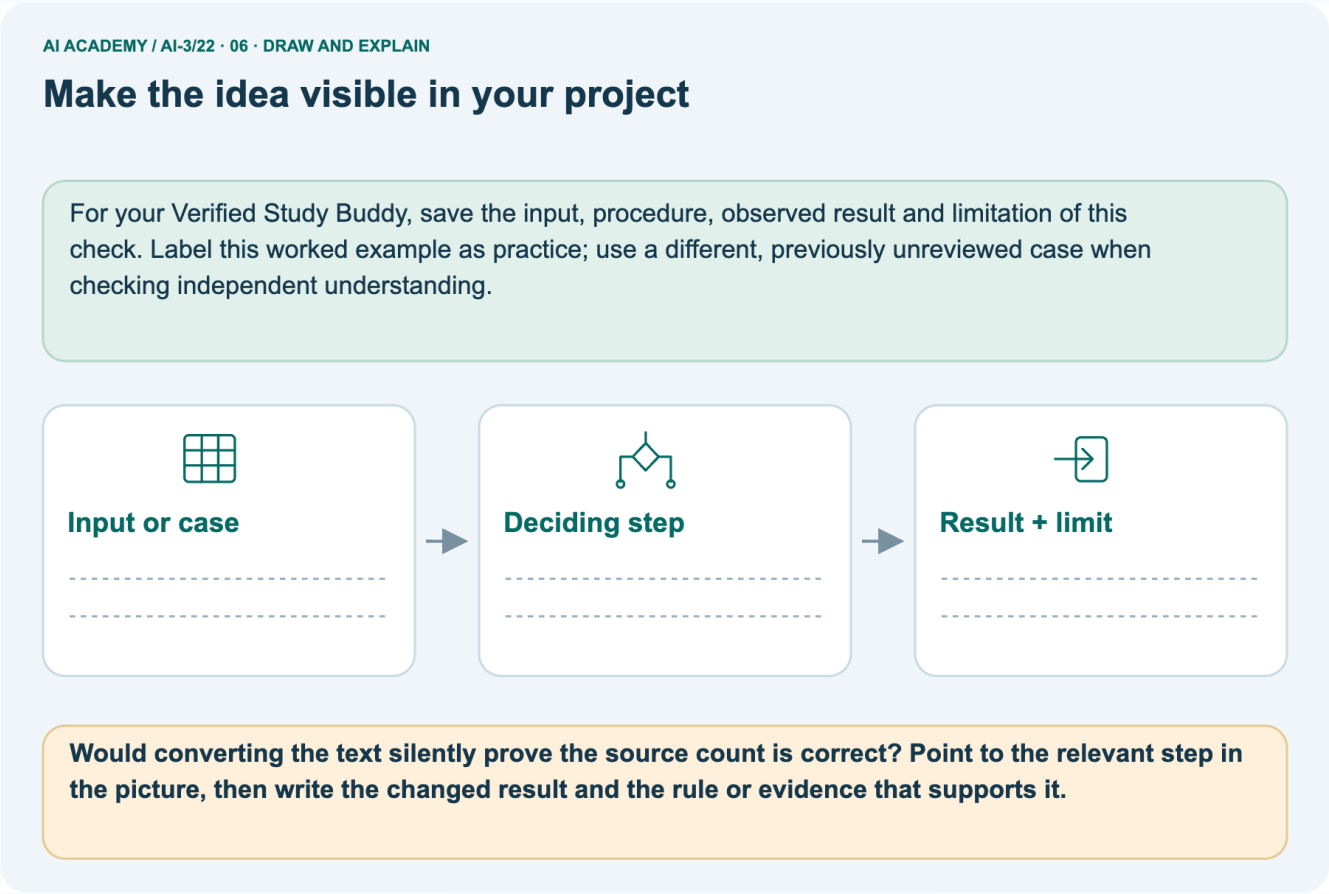
Read the picture in order. Accept this **response** shape; acceptance does not prove that 12 portions were actually counted. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: **JSON** well-formedness, schema validity and real-world truth are different checks.

Explain the case, then point to a limit on the claim.

### My picture explanation, deciding evidence and one limit

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Would converting the text silently prove the source count is correct? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

**My picture explanation, deciding evidence and one limit**

▼ Earlier wording, alternatives and picture transcripts

**01. Requests Responses and JSON: the case**

**02. Follow the deciding steps**

**03. Follow the evidence**

**04. Change one thing. Trace what follows.**

**05. Connect the explanation to its limit**

**06. Make the idea visible in your project**

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.